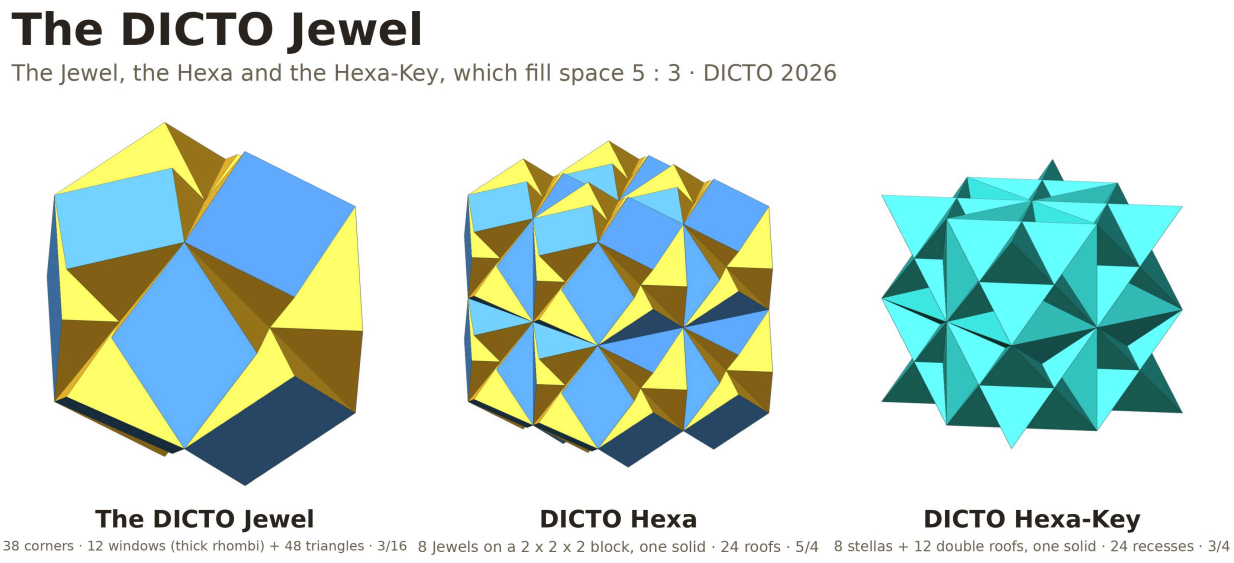
The DICTO Jewel

Euclid's dodecahedron carved by its neighbours' stella octangulas: an exact solid with twelve Penrose windows, its lattice, the DICTO Jewel-Tetra, Jewel-Octa and Jewel-Rhombi and how they fill space, and the DICTO Hexa and Hexa-Key, all exact

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Version 1 · 20 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/d475b59/studies/2026-10-20-dicto-jewel (research branch, commit d475b59; also attached to this record) · Interactive: dictospheres.com → Kaleidohedra → DICTO → Stella–Jewel Lattice
The Jewel, its stella-octangula counterpart, its name, its clusters (and the Jewel-Tetra, Jewel-Octa and Jewel-Rhombi as single solids, and their names), the Hexa, the Hexa-Key and their names are DICTO's (findings #10, #10b, #15 and #18 on dictospheres.com); computation, exact proofs and drafting with Claude (Anthropic).



Abstract. Put Euclid's dodecahedron on a cube, as roofs on its six faces, and carve away the stella octangulas of the six neighbouring cubes: what remains is the **DICTO Jewel**, with 38 corners and 60 faces, twelve of them thick Penrose rhombi (the windows) and 48 triangles, and volume exactly 3/16 at the scale where a 2 × 2 × 2 block of cells (the Hexa's cube, §4) has edge 1, or 12 on cells of edge 2. Jewels and stella octangulas on the other cells fill space in the rock-salt pattern, 3 : 1 by volume [2]; we prove this exactly: each stella's cube is the stella and six carved roofs of exactly 1/96 each. DICTO's clusters [2, 10], named here as single solids: the **DICTO Jewel-Tetra** (4 Jewels, 228 faces, 3/4), whose copies share out all Jewels, and the **DICTO Jewel-Octa** (6 Jewels and the stella that exactly fills their hole, 19/16), which tiles space with *loose* stellas (stellas inside no piece) on exactly 12 lattices of translations, all one tiling up to its symmetry. Together they sit in the pattern of the octet truss, eight Jewel-Tetras round each Jewel-Octa, but as overlapping solids, not as a tiling: without overlap at most one stella in six can be inside a Jewel-Octa, and the 12 lattices reach that bound. The third, the **DICTO Jewel-Rhombi**, is

eight Jewels on the corners of the face-centred cubic cell, meeting on 18 windows, with the stella sealed inside (396 faces, 25/16). Each of the three tiles space with loose stellas (on 4, 12 and 25 lattices of translations; 2, 1 and 9 up to symmetry); none can without stellas, since Jewels are only three quarters of space. The **DICTO Hexa** and **DICTO Hexa-Key** [12] (5/4 and 3/4, filling space 5 : 3, two windows between neighbouring Hexas), checked numerically in [12], are proved exactly. As a solid the Jewel is pyritohedral, so it twins like the Ogdoecontahedron [5]; its packing does not twin by a (111) mirror, because that mirror does not keep the cube lattice. All claims are exact in $\mathbb{Q}(\sqrt{5})$ except where marked float.

1. The Jewel (exact)

Frame. As in the EKP cell [1] and [2]: cells are cubes centred on a cubic lattice; a cell is *even* or *odd* by the parity of its index sum, and the even cells form a face-centred cubic lattice. All values below are at the scale where the Hexa's cube (§4) has edge 1, so a cell has edge 1/2 and volume 1/8; the scripts use cells of edge 2, where every volume is 64 times larger.

Construction. Euclid builds the regular dodecahedron on a cube by setting a roof on each face [3]: the hull of the face and a ridge of length $1/\varphi$ times the cell edge, at height $(\varphi - 1)/2$ times the cell edge; each roof is $(1 + \sqrt{5})/192$ and the dodecahedron $(5 + \sqrt{5})/32$. Each roof reaches into the next cell, where the stella octangula (Kepler's two tetrahedra on the cube's corners [4], volume 1/16) takes away $(\sqrt{5} - 1)/192$ of it, leaving a **carved roof** of exactly 1/96. The Jewel is its cell and its six carved roofs: $1/8 + 6/96 =$ **3/16**.

Its surface. Each carved roof is exactly four tetrahedra, one over each quarter of the cube face, meeting at the face centre: each has half a window on its outside and two walls in the planes of the neighbouring stella (proved exactly: the four lie in the roof, miss the stella, meet only in faces and add up to 1/96). So the Jewel has 38 corners: the cube's 8, the 12 ridge points, the 6 face centres (where it is dimpled), and 12 points that complete the windows. Its 60 faces are 12 **windows**, one across each cube edge, lying in the dodecahedron's face planes, and $6 \times 8 = 48$ triangles in the stellas' planes; $38 - 96 + 60 = 2$. Each window is a thick Penrose rhombus: side $(\sqrt{5} - 1)/4 = 1/(2\varphi)$, long diagonal 1/2 (the cell's edge), angle 108° at the ridge corners ($\cos 108° = (1 - \sqrt{5})/4$, exact) and 72° at the cube corners [6]. The Jewel's corners are centrosymmetric, have the three-fold turns and the three coordinate mirrors, but no diagonal mirror and no four-fold turn: its symmetry group has order 24, and it is pyritohedral, like Euclid's roofed cube itself. It also has no mirror across a (111) plane (no cubic solid has), which §5 uses.

2. The Stella–Jewel Lattice (exact)

Jewels on the even cells and stella octangulas on the odd cells fill space, 3/16 : 1/16 = 3 : 1 [2]. The first check (finding #10b) sampled points; here it is exact. Take an odd cell. The six Jewels on its face-neighbours each reach into it with one carved roof (their ridges are Jewel corners). These six roofs lie in the cell, any two meet only in a face (the intersection of the uncarved roofs has volume exactly 0), and each carved part is 1/96. With the stella they give $1/16 + 6/96 = 1/8$, the cell: so an odd cell is exactly its stella and six carved roofs, an even cell is the core of its Jewel, and the Jewels and stellas tile space.

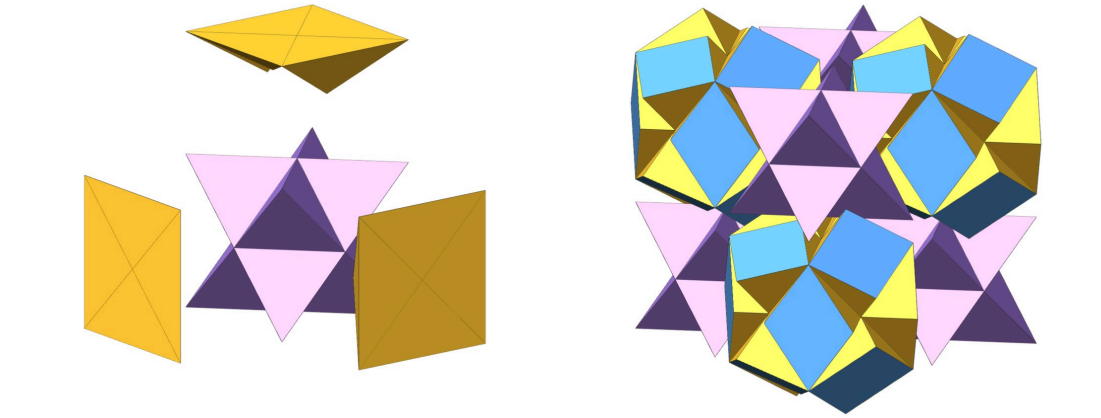


Figure 1. Left: an odd cell opened, its stella octangula (violet) and three of the six carved roofs that the neighbouring Jewels push into it. Right: a 2 × 2 × 2 block of the Stella–Jewel Lattice, drawn slightly apart.

3. The Jewel-Tetra, Jewel-Octa and Jewel-Rhombi (exact)

Neighbours. Two Jewels on neighbouring even cells (one step along a face diagonal) share exactly one window; Jewels two cells apart along an axis share none.

The DICTO Jewel-Tetra (DICTO's tetrahedral cluster, finding #15, as one solid): four Jewels on mutually neighbouring even cells. Each pair shares one window, so its surface has (counted) 4 × 60 − 2 × 6 = **228 faces** (192 triangles, 36 windows), and all four meet at one point, their cells' common corner; volume 3/4. Every set of four mutually neighbouring Jewels is of this kind, one per cell corner. Anchored on the cells whose three indices are all even, Jewel-Tetras use every Jewel exactly once, with the stellas loose between them.

The DICTO Jewel-Octa (DICTO's octahedral cluster, finding #15, as one solid): the six Jewels round an odd cell, and that cell's stella. Each Jewel shares a window with its four neighbours in the ring (12 windows); inside, the 48 walls of the Jewels' carved roofs enclose a stella-shaped hole, sealed but for its eight tips, and since the odd cell is exactly its stella and the six carved roofs (§2), the stella fills the hole exactly. Since Jewels meet each other only on whole windows and every Jewel triangle lies on a stella, the faces can be counted: the solid has an outer surface of **288 faces** (240 triangles, 48 windows; 6 × 60 − 2 × 12 − 48) and volume 6 × 3/16 + 1/16 = **19/16**. Jewel-Octas tile space together with *loose* stellas (stellas inside no piece) on exactly **12 lattices of translations of index 12** (a lattice of index n holds one cell in n; exact integer search over all lattices of that index), for example the one with Hermite basis (1,0,3), (0,1,5), (0,0,12) in cell units; each period holds one Jewel-Octa and five loose stellas. Up to the Jewel-Octa's own symmetry the 12 are one tiling.

The octet pattern (DICTO's octet network [10]). The Jewels' cells form a face-centred cubic lattice; its tetrahedral holes are the cell corners and its octahedral holes the odd cells, so Jewel-Tetras and Jewel-Octas sit exactly where the octet truss has its tetrahedra and

octahedra. Each Jewel lies in 8 Jewel-Tetras and 6 Jewel-Octas (counted), and each of a Jewel-Octa's 8 faces (three mutually neighbouring Jewels) is completed to a Jewel-Tetra by exactly one Jewel outside. It is not an octet tiling: neighbouring Jewel-Tetras and Jewel-Octas share Jewels, and what fills space is still the Jewels and the stellas. Without overlap it cannot be one either: each Jewel-Octa takes six Jewels and one stella, and there are as many Jewels as stellas, so at most one stella in six can be inside a Jewel-Octa; the 12 lattices above reach that bound, leaving five stellas in six loose.

The DICTO Jewel-Rhombi (the rhombohedral cluster of [2], as one solid): eight Jewels on the corners of the face-centred cubic cell, which is the Jewel-Octa with the two Jewels at the ends of its long diagonal. Its faces have 60° angles, so the Jewels meet not only along its 12 edges but also along the 6 short face diagonals: **18 windows**. All six face-neighbours of the odd cell at its centre are among its corners, so that cell's stella is sealed inside and fills the hole exactly, as in the Jewel-Octa. Counted in the same way, as one solid it has **396 faces** (336 triangles, 60 windows; 8 × 60 − 2 × 18 − 48) and volume 8 × 3/16 + 1/16 = **25/16**, 8⅛ Jewels.

How they fill space. The fillers are always the Jewels and the stellas; the pieces are ways of grouping them. In the Stella–Jewel Lattice there are as many stellas as Jewels, and Jewels are only three quarters of space, so no pieces made of Jewels alone can fill it: the stellas have to come too. Each of the three pieces fills space with copies of itself and *loose* stellas (stellas not inside any piece) on a lattice, and we found every such lattice of translations (exact integer search over all lattices of the right index; a period holds as many stellas as Jewels, so the index is twice the Jewels per piece). Many are the same tiling turned; the last column counts them up to the piece's own symmetry:

piece	Jewels	stella inside	loose stellas per piece	lattices	up to symmetry
Jewel-Tetra	4	0	4	4 (index 8)	2
Jewel-Octa	6	1	5	12 (index 12)	1
Jewel-Rhombi	8	1	7	25 (index 16)	9

The Jewel-Octa takes in the most stellas, one for every six Jewels, which is the bound of the octet pattern above. Grouped with sharing instead, the pieces make networks: Jewel-Tetras sharing single Jewels corner to corner form a pyrochlore network (DICTO's Kagome network, finding #16 [10]; corner-sharing tetrahedra are the classical pyrochlore lattice), and Jewel-Tetras with Jewel-Octas form the octet pattern.

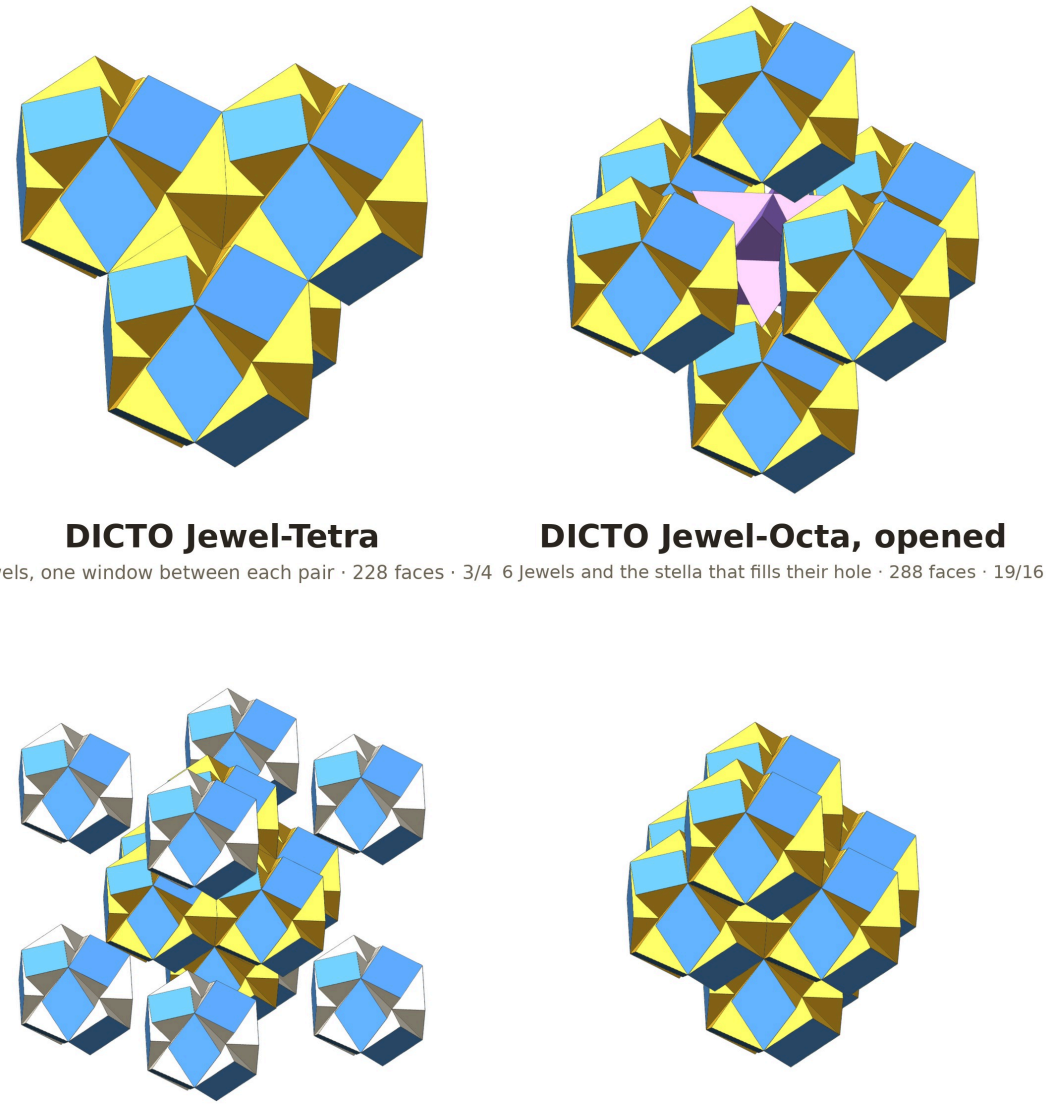


Figure 2. The DICTO Jewel-Tetra; the DICTO Jewel-Octa, its Jewels drawn out to show the stella (violet) that fills its hole; the octet pattern, a Jewel-Octa (gold) and the 8 Jewels (grey) that complete its faces to Jewel-Tetras, drawn out; and the DICTO Jewel-Rhombi (windows blue).

4. The DICTO Hexa and the DICTO Hexa-Key (exact)

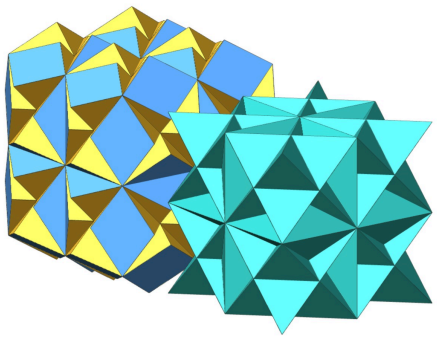
DICTO's Hexa and Hexa-Key, their 5 : 3 filling and their 24 windows are published in [12], where the volumes and windows were checked numerically; here they are proved exactly.

The Hexa. Put a whole Jewel on each of the eight cells of a 2 × 2 × 2 block, on the even and the odd cells alike. Inside the block they overlap; outside it, each of the 24 outer cell faces carries its Jewel's roof. As one solid, the **DICTO Hexa** is the block (edge 1) with 24 carved roofs on its faces: 1 + 24/96 = **5/4**, six and two-thirds Jewels.

The Hexa-Key. Put Hexas on alternate blocks, rock-salt fashion (the blocks of even index sum). Each other block is then entered by 24 roofs, four through each face, from its six neighbouring Hexas; what is left is one solid, the **DICTO Hexa-Key**: $1 - 24/96 = \mathbf{3/4}$, the volume of four Jewels. It is eight stella octangulas, one per cell, joined by twelve double roofs across the inner cell faces: $8/16 + 12 \times 2/96 = 3/4$. Its 24 recesses are exactly the Hexas' 24 roofs, so Hexas and Hexa-Keys fill space, **5 : 3**.

The same pattern one level up [12]. A cell with six carved roofs added is a Jewel ($1/8 + 6/96$), with six taken away a stella ($1/8 - 6/96$); a block with 24 carved roofs added is a Hexa, with 24 taken away a Hexa-Key; and both levels sit in the rock-salt pattern [7].

Windows. The Hexas sit on a face-centred cubic lattice of blocks. Each Hexa meets each of its 12 neighbours on exactly **two whole windows** (24 in all, exact), and no two Hexas' roofs overlap. The Hexa is a union of overlapping Jewels, which is why it is defined as one solid.



Hexa and Hexa-Key
the Hexa's roofs fit the Hexa-Key's recesses · 5 : 3

DICTO · dictospheres.com

Figure 3. A Hexa and the Hexa-Key next to it in their 5 : 3 filling, drawn out a little: the Hexa's roofs go into the Hexa-Key's recesses.

5. Twins by a (111) mirror

By the twin rule [5], a centrosymmetric solid with a three-fold axis has as its mirror image in a plane at right angles to that axis the same solid turned 60°. The Jewel is such a solid, so as a solid it twins like the DICTO Ogdoecontahedron. Its packing does not twin by a (111) mirror: in [8] such a twin through a (111) layer of the Stella–Jewel Lattice left about 4% of sampled points uncovered and 4% covered twice (float). The reason is exact and simple: the mirror in a (111) plane takes the cubic lattice of cells to a different lattice (the step to the next cell, (1, 0, 0) in cell units, goes to $(1/3, -2/3, -2/3)$), so the mirrored packing cannot continue the cubic cells across the plane. The Ogdoecontahedron's packing twins because its cells, octahedra, are mirrored into the next octahedra of the gyrated octet; the Jewel's cells are cubes, and a mirrored cube does not continue the cubic lattice.

6. Prior art and what is claimed

- **Classical parts.** The dodecahedron as a cube with six roofs is Euclid's (Elements XIII.17) [3]; the stella octangula is Kepler's [4]; the thick rhombus is Penrose's [6]; the rock-salt arrangement is that of sodium chloride [7]; the EKP cell holds them together [1].
- **Lalvani.** US 4,723,382 [9] builds periodic and non-periodic structures from polygonal members whose edges all lie on the fifteen two-fold axes of the icosahedral group. All 48 window edges of the Jewel lie on those axes, but none of its other 48 edges do (lalvani_edges.py, float), so the Jewel, and with it the stella and the lattices here, fall outside that edge condition (our reading, not legal advice).
- **Already published.** The Jewel's definition, its volume and the Stella–Jewel Lattice are in [2], with the cluster networks; [2] also gives the octahedral cluster with its stella and the rhombohedral cluster as closed solids with their volumes; the Jewel, the lattice, the tetrahedral and octahedral clusters and the octet network were first released with the Kaleidohedra app (findings #10, #10b, #15 [10]); the Hexa, the Hexa-Key, their 5 : 3 filling and the two windows between Hexas are in [12], which checked their volumes and 24 windows numerically.
- **Claimed here:** the exact proofs (in $\mathbb{Q}(\sqrt{5})$) of the Jewel's corners, windows, surface, volume and symmetry and of the Stella–Jewel tiling; for the three cluster solids of [2, 10], here named the DICTO Jewel-Tetra, Jewel-Octa and Jewel-Rhombi, their face counts, the 18 windows of the Jewel-Rhombi and the proof that the stella fills the hole exactly; all lattice tilings of the three with loose stellas (4, 12 and 25; 2, 1 and 9 up to symmetry); the octet pattern and its bound of one stella in six; the exact proofs of the Hexa, the Hexa-Key, their 5 : 3 filling and the two-window contacts, checked numerically in [12]; and the exact reason the packing does not twin by a (111) mirror.

In the sources checked (the full text of the 80 library sources listed in the appendix, and web searches on 20 October 2026) we did not find the Jewel, its clusters, the Hexa or the Hexa-Key outside DICTO's own records [2, 10, 12].

7. Further work

- A third level: Hexas on a 2 × 2 × 2 block of blocks.
- A physical puzzle: Hexas and Hexa-Keys interlock and can be cut in wood or printed.

Data and reproducibility

Exact ($\mathbb{Q}(\sqrt{5})$): jewel_exact.py (roof, Jewel and its four-tetrahedron roofs, symmetry, Stella–Jewel tiling, Hexa and Hexa-Key, windows between Hexas, the (111) mirror; log jewel-exact.log, 21 PASS) and cluster_exact.py (Jewel-Tetra, Jewel-Octa and Jewel-Rhombi, their lattice tilings and classes up to symmetry, the octet pattern and its bound; log cluster-exact.log, 24 PASS); both run with the exact tools of [11] (python3 <script> studies/2026-10-18-ocd-octet/scripts). Float: lalvani_edges.py (python3 lalvani_edges.py studies/2026-10-19-og-gyrated-octet, which holds jewel_twin.py); the face counts 228, 288 and 396 are counted from the exact shared windows and stellas. Figures: jewel_figs.py (float copies of the exact pieces), shots.mjs (2× pixel ratio), jewel_compose.py.

References

1. DICTO, The Euclid–Kepler–Pacioli Cell, DICTO DOI: 10.5281/zenodo.23284106.
2. DICTO, Kagome Hulls, DICTO DOI: 10.5281/zenodo.23376317.
3. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III (Books X–XIII), Cambridge (1908), Book XIII, Proposition 17.
4. J. Kepler, Harmonices mundi libri V (1619), Book II.
5. DICTO, The DICTO OG Family on Every Stacking, DICTO DOI: 10.5281/zenodo.23490114.
6. R. Penrose, Pentaplexity: a class of non-periodic tilings of the plane, Mathematical Intelligencer 2, 32–37 (1979).
7. W. L. Bragg, The structure of some crystals as indicated by their diffraction of X-rays, Proc. R. Soc. Lond. A 89, 248–277 (1913).
8. DICTO, The DICTO Ogdoecontahedron on the Gyrated Octet, DICTO DOI: 10.5281/zenodo.23484012.
9. H. Lalvani, Building structures based on polygonal members and icosahedral symmetry, US Patent 4,723,382 (1988).
10. DICTO, Kaleidohedra by DICTO, DICTO DOI: 10.5281/zenodo.23173809; releases 10.5281/zenodo.23223555 (#10b) and 10.5281/zenodo.23257393 (#15, #16).
11. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016 (the exact tools q5.py and exactpoly.py).
12. DICTO, The DICTO Hexa, DICTO DOI: 10.5281/zenodo.23376320.

Appendix. The reference library searched

The full text of the 80 sources below in DICTO's reference library was searched on 20 October 2026 for: Penrose, stella octangula, rock salt, Lalvani, Euclid, roof and rhombic. None contains the results of this paper. The relevant hits are Euclid and Kepler [3, 4], Pacioli, Messer's stellations of the rhombic triacontahedron (the stella octangula), Lalvani's NASA report on octahedral–tetrahedral trusses, and the papers on Penrose tilings (Kramer, Baake and Grimm, Dietl and Eschenburg). Thirteen further items are page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, among them US 4,723,382, read separately in full on 8 October 2026, and Fedorov 1885).

Web searches (20 October 2026): a dodecahedron carved by stella octangulas; space-filling pairs of a dodecahedron-based solid and the stella octangula; cubes of overlapping dodecahedra.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra

S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
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